D121 | Forecast accuracy is not order authority

PRIVATE FICTIONAL one-SKU, one-week-ahead exercise. No real festival calendar, demand data, replenishment, supplier contact, inventory change, rollout or spend.

Attempt first	Rebuild the ledger and all three error/cost tables before viewing references. Record an actual response only if performed. No model fitting, observed learner mastery or empirical deployment result is claimed.
Frozen common comparison	Four separate rolling origins issue one-week forecasts. Demand is fully observed with adequate stock in these four windows. A is a locked prior-season festival-relative baseline, not a same civil-date rule. B/C were fixed before these outcomes.
Baseline A	Demand 100/120/80/140; A 90/110/100/130. Errors actual-minus-forecast10/10/-20/10. MAE 12.5, MSE 175, RMSE 13.228757, mean error +2.5. Shortage 30/excess 20 costs 140 at 4/1 rupees per unit.
Ranking disagreement	B 100/120/100/140 has RMSE 10 and cost 20. C 100/120/80/130 has RMSE 5 but cost 40. Smaller RMSE is not necessarily smaller asymmetric cost. Point forecasts are not automatically order quantities.
Time and features	Ledger uses synthetic ordinal weeks, not calendar dates. Feature as-of must be no later than issue; target minus issue must equal the required horizon. These checks do not inspect model training or prove all features truly existed.
Separate censored sale	Availability 90/sales 90/known stockout: demand>=90, exact demand unknown. Forecast 110 implies error>=-20 with no finite upper bound, not verified error-20. Adequate availability 150/sales 90/no stockout identifies demand 90 under this simple model.
Limits and authority	Static prose is Case-v1 and needs revision after edits. Loss assumes separate windows, no carryover/salvage/substitution/backorders. No new empirical/full AV/cross-viewer/independent learner/account clearance or overall10. Real orders require separate authority.

Inputs | Frozen demand and point forecasts

Four equally weighted, one-week windows, one SKU. Quantities are units; costs are INR per unit. Yellow inputs are fictional. A repeats corresponding prior-season festival-relative weeks.

Window	Demand	Seasonal A	Candidate B	Candidate C
Window 1	100.00	90.00	100.00	100.00
Window 2	120.00	110.00	120.00	120.00
Window 3	80.00	100.00	100.00	80.00
Window 4	140.00	130.00	140.00	130.00

Shortage INR/unit	4.00
Excess INR/unit	1.00

Structural quantities/costs valid? (1=yes)	1.000000

All four demands are stipulated fully observed. Candidates are supplied frozen outputs, not estimated by this workbook. Missing/text/negative inputs hold comparisons. Zero cost is allowed, not evidence that a real cost is absent.

Ledger | Issue time, target and available features

Ordinal synthetic weeks only. No actual calendar commitments. Horizon is target minus issue. Equal row definitions across all three methods; timestamps cannot validate unseen training pipelines.

Window	Issue week	Target week	Feature as-of	Horizon	Row valid
Window 1	5.00	6.00	5.00	1.00	1.000000
Window 2	6.00	7.00	6.00	1.00	1.000000
Window 3	7.00	8.00	7.00	1.00	1.000000
Window 4	8.00	9.00	8.00	1.00	1.000000

Required horizon	1.00

Ledger structurally aligned? (1=yes)	1.000000

Changing required horizon to4 alone must HOLD the one-step comparison. A valid four-step ledger also needs genuinely frozen four-step forecasts and matching outcomes, not just edited labels. Do not mix a model-selection backtest with an untouched final holdout.

Seasonal A | Same outcomes, different losses

Error = actual demand minus forecast. Positive is underforecast; shortage/excess are descriptive forecast deviations, not an executed stock policy. All forecasts precede outcomes by stipulation.

Window	Signed error	Absolute	Squared	Shortage	Excess	Toy loss INR
Window 1	10.000000	10.000000	100.000000	10.000000	0.000000	40.000000
Window 2	10.000000	10.000000	100.000000	10.000000	0.000000	40.000000
Window 3	-20.000000	20.000000	400.000000	0.000000	20.000000	20.000000
Window 4	10.000000	10.000000	100.000000	10.000000	0.000000	40.000000

Units: signed/absolute/shortage/excess in SKU units, squared errors in units squared, loss in INR. Do not add errors across a different horizon, SKU or outcome definition. Zero mean error can conceal large opposing errors.

Candidate B | Same outcomes, different losses

Error = actual demand minus forecast. Positive is underforecast; shortage/excess are descriptive forecast deviations, not an executed stock policy. All forecasts precede outcomes by stipulation.

Window	Signed error	Absolute	Squared	Shortage	Excess	Toy loss INR
Window 1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Window 2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Window 3	-20.000000	20.000000	400.000000	0.000000	20.000000	20.000000
Window 4	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

Units: signed/absolute/shortage/excess in SKU units, squared errors in units squared, loss in INR. Do not add errors across a different horizon, SKU or outcome definition. Zero mean error can conceal large opposing errors.

Candidate C | Same outcomes, different losses

Error = actual demand minus forecast. Positive is underforecast; shortage/excess are descriptive forecast deviations, not an executed stock policy. All forecasts precede outcomes by stipulation.

Window	Signed error	Absolute	Squared	Shortage	Excess	Toy loss INR
Window 1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Window 2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Window 3	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Window 4	10.000000	10.000000	100.000000	10.000000	0.000000	40.000000

Units: signed/absolute/shortage/excess in SKU units, squared errors in units squared, loss in INR. Do not add errors across a different horizon, SKU or outcome definition. Zero mean error can conceal large opposing errors.

Summary | Error ranking can disagree with cost

Equal weights over four fixed windows; no significance test or deployment guarantee. Lower is better for MAE/RMSE/cost, but signed error is bias direction, not a quality ranking.

Metric	Seasonal A	Candidate B	Candidate C
Mean signed error (units)	2.500000	-5.000000	2.500000
Mean absolute error (units)	12.500000	5.000000	2.500000
Mean squared error (units squared)	175.000000	100.000000	25.000000
Root mean squared error (units)	13.228757	10.000000	5.000000
Total shortage deviation (units)	30.000000	0.000000	10.000000
Total excess deviation (units)	20.000000	20.000000	0.000000
Total toy inventory loss (INR)	140.000000	20.000000	40.000000

B minus C RMSE, units	5.000000
B minus C toy loss, INR	-20.000000
Shortage-cost B/C tie, if identified	2.000000

Case-v1 B/C loss ties at shortage cost 2 when excess cost 1, because B has 20 excess and C 10 shortage. The formula recomputes the tie from current deviations. Equal shortage deviations give UNDEFINED: costs tie for all shortage costs if excess costs also match, otherwise never. Negative thresholds are infeasible at nonnegative costs. No actual procurement price or universal inventory policy inferred.

Censoring | Sales are not always demand

Separate lost-sales/no-backorder observation with reliable counts, no returns/substitution/replenishment. Stockout=1 means demand may exceed availability; exact lost sales not supplied.

Available units	90.00
Observed units sold	90.00
Reliable known-stockout indicator (0/1)	1.00
Frozen forecast units	110.00

Censoring inputs structurally valid? (1=yes)	1.000000

Demand lower bound, units	90.000000
Exact demand, if identified	UNKNOWN
Demand-error lower bound, units	-20.000000
Exact demand forecast error, if identified	UNKNOWN
Demand-error upper bound	UNBOUNDED

Baseline demand>=90 and error>=-20; exact demand/error UNKNOWN, upper error UNBOUNDED. Availability150/sales90/confirmed no stockout instead gives demand 90/error-20. A missing or unreliable stockout flag is not0. This sheet does not infer an algorithm for recovering lost demand.

Audit | Identity checks are not release authority

Forecast and censoring cases are independent. Broken references hold the affected summary. Ledger checks cover supplied numeric relations, not unseen feature history or actual forecasting model execution.

Forecast identities valid? (1=yes)	1.000000
Guarded B minus C toy-loss contrast INR	-20.000000

Censoring identities valid? (1=yes)	1.000000
Guarded demand-error lower bound units	-20.000000

Passing specified identities does not prove all possible spreadsheet faults absent, actual feature availability, real inventory records, statistical superiority, independent learning, supplier capacity or permission to place an order. Actual review stays blank.

Evidence | Past-only, horizon-matched, uncensored

Selected sources rechecked13 November 2026; no full video, full slide deck, full censored-demand paper, empirical dataset, real participant or model training claimed.

Rolling-origin text	https://otexts.com/fpptr/tscv.html - prior-only training, rolling forecast origins and operationally matching multi-step horizons. Source RMSE model-selection suggestion is not a universal inventory-loss rule.
Selected evaluation slides	https://robjhyndman.com/files/1-ForecastEvaluation.pdf - seasonal-naive definition and selected error-metric teaching. Historical plots and complete deck are not newly visually reviewed. No historical stock/visitor value is a current forecast.
Publisher abstract only	https://pubsonline.informs.org/doi/10.1287/msom.2015.0520 - sales censored by inventory, inventory-record uncertainty can bias demand estimates. Abstract does not supply a general exact lost-demand correction.
Original case and changes	Original D121 synthetic demand/forecast/cost/censoring case retained at https://top1os.dev/. Ledger week numbers are added fictional metadata, not recovered actual issue timestamps. No combined score across independent cases.
Leakage objection	Fit transforms/features/tuning only within past folds. Exogenous features must actually exist at issue. Freeze rules before this test. A backtest used to select a model is not also an untouched final test. Confirm calendar/festival alignment.
Operational objection	Independent-window loss ignores carryover, service goals, lead-time distribution, capacity, salvage, substitutions and stock uncertainty. Four descriptive errors do not establish future policy performance or significance.
Day0/day30 hold	Forecast/inventory/cash/governance are fictional roles, not accepted assignments. Require permitted evidence and separately approved order before action. Actual response, correction effect, reviewer objection and delayed transfer remain blank.

Actual work | Record only what occurred

Actual learner attempts, confidence, errors, reviewer objections and delayed changed-context performance remain blank.

Actual case/date	Response/confidence	Error/reason	Changed input	Correction/effect	Objection/reply	Delayed result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Independent baseline formula checks	PASS: all112 baseline formulas independently recomputed, including text unknowns.
Demand/forecast zero/changed values	PASS:20 changed sets with demand/forecast/zero/fractional forecasts.
Cost-ranking reversal and tie	PASS:cost reversal, tie, zero costs and recomputed tie boundary.
Issue/as-of/target/horizon checks	PASS:as-of, target, horizon and missing-ledger failures held.
Censoring vs adequate-stock changes	PASS:censored/no-stockout/zero-availability demand cases.
Invalid inputs / independent isolation	PASS:18 invalid cases held affected summary, independent case preserved.
Broken result references / holds	PASS:6 broken result references held affected Audit, other case preserved.
Restored baseline / actual work blank	PASS:baseline restored; actual learner fields remain blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.